

PETROZAVODSK, Russian Federation

WMO: 228200

Lat: 61.8164N

Lon: 34.2550E

Elev: 364

StdP: 14.50

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-11.0	-21.3	1.7	-16.7	-14.8	2.5	-10.8	19.8	27.0	17.9	21.5	2.9	240	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	80.0	66.6	76.4	64.6	73.1	62.5	68.6	76.9	66.3	73.7	64.2	71.0	6.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	95.1	73.3	63.2	88.0	70.2	61.3	82.2	68.0	33.0	77.2	31.1	73.7	29.6	71.0	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.9	13.3	-21.4	84.5	7.0	4.1	-26.4	87.4	-30.5	89.9	-34.4	92.2	-39.4	95.2		
(5)			-21.6	71.1	6.9	3.0	-26.6	73.2	-30.6	75.0	-34.5	76.6	-39.5	78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	16.0	16.0	25.0	36.8	47.9	57.1	62.9	59.2	50.3	39.2	28.6	21.5		
	DBStd	18.69	13.33	13.45	9.37	7.98	8.68	7.36	6.32	6.13	6.37	7.19	8.73	11.77		
	HDD50	5291	1055	952	775	404	145	17	2	4	72	341	642	882		
	HDD65	9759	1520	1372	1240	847	531	257	115	198	440	800	1092	1347		
	CDD50	1094	0	0	0	7	81	229	401	289	82	5	0	0		
	CDD65	87	0	0	0	0	1	19	49	18	0	0	0	0		
	CDH74	579	0	0	0	0	30	140	303	105	1	0	0	0		
	CDH80	96	0	0	0	0	7	23	51	15	0	0	0	0		
	WSAvg	6.3	6.2	6.4	7.0	6.6	6.6	6.0	5.2	5.4	5.6	6.6	6.7	7.0		
	PrecAvg	22.6	1.2	0.9	1.2	1.4	1.7	2.3	2.7	3.2	2.7	2.3	1.7	1.5		
	PrecMax	30.9	2.0	1.7	2.7	3.9	3.7	4.7	6.7	7.4	7.4	4.9	3.5	3.7		
(15) Precipitation	PrecMin	15.0	0.1	0.2	0.4	0.2	0.1	0.5	0.2	1.3	0.6	0.0	0.2	0.5		
	PrecStd	3.8	0.4	0.4	0.5	0.8	0.7	1.2	1.4	1.5	1.5	1.0	0.7	0.7		
	DB	37.1	38.5	50.1	64.4	78.1	82.9	85.2	82.0	71.0	58.2	46.7	43.8			
	MCWB	35.7	36.4	40.4	51.2	58.8	66.9	69.0	68.9	61.1	53.1	45.0	42.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	35.1	36.0	42.9	57.9	71.1	77.8	80.6	76.8	65.9	53.8	44.5	39.1			
	2% MCWB	34.1	34.6	36.2	47.6	56.3	64.5	67.8	65.6	58.2	50.8	42.9	38.0			
	5% DB	33.7	34.4	39.0	53.3	66.3	73.9	77.3	73.2	62.4	51.3	42.4	36.6			
	5% MCWB	32.9	33.4	34.6	44.8	53.8	61.9	66.0	63.1	56.8	48.9	41.2	35.7			
	10% DB	32.2	32.8	36.2	48.5	62.4	70.0	73.7	69.7	59.4	49.2	40.1	34.7			
	10% MCWB	31.5	31.9	33.2	41.4	51.8	59.8	63.8	61.4	55.1	47.2	39.1	33.8			
	0.4% WB	35.8	36.9	40.9	52.9	62.7	69.9	72.1	70.3	62.6	54.1	45.3	42.0			
	0.4% MCDB	36.3	37.8	48.5	62.6	73.3	79.0	81.3	78.5	68.3	56.5	46.1	43.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.1	34.8	37.4	48.5	58.4	66.7	69.4	66.6	59.6	51.3	42.9	38.1			
	2% MCDB	34.6	35.6	41.7	56.2	65.9	75.4	78.0	74.5	64.0	53.5	44.0	38.7			
	5% WB	32.9	33.4	35.1	45.5	55.4	63.7	67.2	64.4	57.5	49.3	41.1	35.6			
	5% MCDB	33.5	34.3	37.6	52.1	63.9	71.3	74.7	71.0	61.3	51.0	42.2	36.2			
	10% WB	31.6	32.0	33.7	42.1	52.6	60.7	65.3	62.3	55.7	47.3	39.0	33.7			
	10% MCDB	32.3	32.8	35.7	47.6	61.2	68.1	71.9	67.9	58.9	49.0	39.9	34.4			
	MDBR	9.6	11.1	12.3	13.8	16.1	15.4	14.5	13.8	11.4	8.1	7.2	8.2			
	5% DB	7.0	8.0	13.9	20.0	23.4	20.7	19.7	19.0	15.8	9.8	7.7	6.5			
(36) Mean Daily Temperature Range	MCDBR	7.2	7.5	9.6	12.2	12.1	10.6	9.8	10.2	9.7	7.3	7.2	6.4			
	MCWBR	6.8	7.9	11.3	17.8	20.1	18.2	16.9	16.4	13.6	8.9	7.4	6.4			
	MCDBR	7.0	7.7	8.5	11.7	11.9	10.5	9.5	9.6	9.1	7.4	7.1	6.5			
	MCWBR															
(40) Clear-Sky Solar Irradiance	taub	0.247	0.296	0.321	0.361	0.356	0.349	0.383	0.368	0.337	0.318	0.264	0.216			
	taud	2.028	2.082	2.152	2.294	2.437	2.483	2.406	2.426	2.496	2.391	2.120	1.970			
	Ebn at Noon	135	204	246	262	275	279	264	258	246	204	129	89			
	Edh at Noon	12	23	31	33	31	31	32	29	23	17	11	7			
(44) All-Sky Solar Radiation	RadAvg	72	259	718	1209	1577	1706	1614	1231	702	268	73	35			
	RadStd	10	58	98	121	128	116	181	113	76	37	12	4			

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.08	N/A	N/A	N/A	N/A	N/A	-427	-412	N/A	N/A
(47) Regional (6 neighbors)		+0.99	N/A	N/A	N/A	N/A	N/A	-400	N/A	N/A	N/A

Nomenclature: See separate page